

Lee Seou

Industrial Engineering

CONTACT

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Sungkyunkwan University
github.com/Seou-Web

SKILLS

Tools

Python MATLAB

Numerical Methods

LU Factorization
Gauss-Seidel Runge-Kutta
FDM FFT
Interpolation

Data Analysis

ANOVA Statistical Analysis

Capstone Design Project

Visualization Simulator for Cosmetic Manufacturing Vessels Using Manufacturing Data

Smart Factory Capstone Design 1 / Team 12

Developed a Population Balance Model (PBM)-based simulator with multiple quality indicators to predict and judge scale-up emulsification quality in cosmetic emulsion manufacturing vessels (CEVSim).

- Responsible for results analysis and project management
- Analyzed droplet size distribution data to evaluate emulsion quality and predict manufacturing outcomes
- Project outcomes resulted in a pending patent application and a registered software work (see Research Outcomes below)

Research Outcomes

Patent Application

Under Examination

- Method and System for Predicting and Determining Scale-up Emulsification Quality of Cosmetic Emulsion Manufacturing Vessels Using a Population Balance Model and Multiple Quality Indicators
- Applicant: Sungkyunkwan University Industry-Academic Cooperation Foundation
 - Inventors: Jeong Jongpil, Kang Mingyu, **Lee Seou**, Lee Hakseung
 - Reference No.: R-2026-0578-KR-1

Software Registration — CEVSim

Registered

- Cosmetic Emulsification Vessel Scale-Up Simulator — Computer Program Work, Application Software / Industrial S/W
- Authors: Jeong Jongpil, Yoo Jaekyung, Kang Mingyu, **Lee Seou**, Lee Hakseung
 - Registration No.: C-2026-030853 · Created May 29, 2026

Research Papers

Scale-Up Simulation of Cosmetic Emulsion Manufacturing Vessels (100/300/500 L) Using a Population Balance Model: Integrated Multi-Metric Analysis of Droplet Size Distribution, Uniformity, Viscosity, and Hardness

Team 12 / Manuscript Submitted

Jeong Jongpil, Yoo Jaekyung, Kang Mingyu, **Lee Seou**, Lee Hakseung — manuscript prepared from the capstone project results, submitted for double-blind peer review (2026).

CFD simulation of a Rushton turbine stirred-tank using open-source software with critical evaluation of MRF-based rotation modeling

Individual / Mar. 2026

Reid, A., Rossi, R., Cottini, C., Benassi, A. — *Meccanica* (Springer), 2025, pp. 1613–1637.

BioSTEAM: A Fast and Flexible Platform for the Design, Simulation, and Techno-Economic Analysis of Biorefineries under Uncertainty

Individual / May 2026

Cortés-Peña, Y. et al. — *ACS Sustainable Chemistry & Engineering*, 2020, 8(8), 3302–3310.

A population balance model for cosmetic emulsion design: A multiscale approach

Team 12 / May 2026

Calvo, F., Gómez, J. M., Ricardez-Sandoval, L., Álvarez, O. — *Chemical Engineering Science*, 2024, 287: 119737.

Education

B.S. in Industrial Engineering — Sungkyunkwan University

2025.03 ~ Present

Lee Seou — Capstone CV

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